

HOMEWORK 5

NUMERICAL ANALYSIS

(MATH 461) FALL, 2023

DUE: 10/20/2023

Problem 1. Suppose $n \geq 2$ and $p \in [1, \infty)$. For $x \in \mathbb{R}^n$, show that $\|x\|_\infty \leq \|x\|_p \leq n^{1/p} \|x\|_\infty$.

Problem 2. Find κ_1 , κ_2 , and κ_∞ for the matrix $A = \begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix}$

Problem 3. The linear system

$$\begin{aligned} 10x_1 + x_2 + x_3 &= 12 \\ x_1 + 10x_2 + x_3 &= 12 \\ x_1 + x_2 + 10x_3 &= 12 \end{aligned}$$

has the unique solution $x_1 = x_2 = x_3 = 1$. Starting with $x^{(0)}$ the zero vector, perform two iterations using Jacobi and one iteration using Gauss-Seidel. Then, calculate the error norms for each of the three iterations. Which method seems to converge faster?

Problem 4. Consider the linear system $Ax = b$. The matrix A is

$$A = \begin{bmatrix} 1.2969 & 0.8648 \\ 0.2161 & 0.1441 \end{bmatrix}$$

with

$$b = \begin{bmatrix} 0.8642 \\ 0.1440 \end{bmatrix}.$$

- (a) An approximate solution is $x = \begin{bmatrix} 0.9911 \\ -0.4870 \end{bmatrix}$. What is the residual vector $r = b - Ax$?
- (b) Does $\|r\| \leq 10^{-5}$ imply $\|e\| \leq 10^{-5}$ (where $\|e\|$ is the error in some vector norm)? Why? What is the true solution?
- (c) Compute the determinant of A , and find the condition number $\kappa_\infty(A)$.

Problem 5. Let A be a 2×2 matrix such that $A_{11} = A_{22} = 1$ and $A_{12} = A_{21} = \alpha \in \mathbb{R}$. Establish a condition on α so that convergence of Jacobi and Gauss-Seidel methods applied to $Ax = b$ with $x^* \neq 0$ is guaranteed.

Problem 6. Consider the matrix

$$A = \begin{bmatrix} 1.1 & 1 & 1 \\ 1 & 1.1 & 1 \\ 1 & 1 & 1.1 \end{bmatrix}$$

- (a) Show that the Jacobi iteration does not converge.

(b) Determine the values for w for which the following iteration converges:

$$x_{k+1} = x_k + w(b - Ax_k).$$

(c) Prove that the Gauss-Seidel iteration converges.

Problem 7. Let α be a scalar and let A be a symmetric positive definite matrix with eigenvalues $0 < \lambda_1 < \dots < \lambda_n$. Consider the iterative scheme

$$x_{k+1} = x_k + \alpha(b - Ax_k).$$

(a) If $A = M - N$ is the splitting associated with this method, state what M and the iteration matrix T are.

(b) Provide, with justification, a condition on α that is sufficient for the iteration to converge for any initialization x_0 .

(c) Show that the optimal choice of α is $\alpha = \frac{2}{\lambda_1 + \lambda_n}$ and the optimal rate of convergence is $\frac{\kappa_2(A) - 1}{\kappa_2(A) + 1}$.

Problem 8. Consider the 2×2 matrix

$$A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

and suppose we are required to solve $Ax = b$.

(a) Write down explicitly the iteration matrices corresponding to the Jacobi, Gauss–Seidel, and SOR schemes.

(b) Find the spectral radius of the Jacobi and Gauss–Seidel iteration matrices and the asymptotic rates of convergence for these two schemes.